

EDUCATION

University of California, Los Angeles

Los Angeles, CA

Bachelor of Science, Mathematics of Computation (Math + CS)

June 2028

- **GPA:** 3.68/4.00
- **Organizations:** Data Resolutions (ML Researcher), Bruin Quant Traders (Analyst)

PROJECTS

DistilBERT Salary Prediction | Python (Scikit-learn, PyTorch, Hugging Face, Pandas, NumPy)

- Fine-tuned a DistilBERT transformer model to analyze 22,000+ job descriptions, allowing the model to understand complex professional context and seniority levels.
- Achieved an R^2 score of 0.69, meaning the model accounts for nearly 70% of the variance in salary based on the job description.
- Achieved a Mean Absolute Error (MAE) of 0.21 on log-transformed data, translating to the model's predictions falling within 24% of actual salary on average.

Quantitative Forecasting Research Project | Python (PyWavelets, Scipy, YFinance)

- Researched Wavelet Decomposition to isolate time-series volatility across multiple frequencies
- Built a multiscale forecasting model by combining Wavelet transformations with ARIMA/GARCH architectures
- Developed a portfolio optimization strategy using Mean-Variance analysis, outperforming standard ARIMA/GARCH on backtested returns
- Generated and presented insights on the benefits of signal denoising, demonstrating that frequency-based decomposition can improve stability of linear forecasting models.

Sound Scout – Music Artist Recommendation Engine | Python, JavaScript, HTML/CSS, Flask

- Built a music recommendation system using TF-IDF to turn artist tags into numerical features
- Used cosine similarity to find artists (21,000+) similar to a user's selected favorites
- Applied K-Means clustering to group artists by style and improve recommendation quality
- Combined cosine similarity scores with cluster matching to return more relevant results

EXPERIENCE

Incoming Business Analyst Intern

Jun. 2026

Capital One

McLean, VA

- Incoming Summer 2026

Machine Learning Researcher

Apr. 2026 - Present

DataRes @ UCLA

Los Angeles, CA

- Applied Machine Learning + Deep Learning Research on topics including reinforcement learning, transformers

Nuclear Operations Training Intern

Jun. 2025 - Sep. 2025

Pacific Gas and Electric (PG&E)

Avila Beach, CA

- Made Power BI dashboard to highlight due dates of SAPNs of Maintenance Training using power query and the SAP SQL database. Dashboard saved ~10 minutes a day for summary of due notifications.
- Automated a long-term schedule for Nuc. Ops. Initial License Training (ILT) using Oracle Primavera P6 with 600+ classroom/simulator events saving ~30 minutes per week over an 18 month term

SKILLS, RELEVANT SOFTWARE & INTERESTS

Skills/Relevant Software: Python, C++, JavaScript, HTML/CSS, Power BI, Tableau

Libraries and Frameworks: Pandas, NumPy, PyTorch, HuggingFace, Matplotlib, XGBoost, RandomForest, scikit-learn,

Interests: NCAA MBB, Card Collecting, Sneakers, Golfing, Board Games, Fantasy Sports